

SECURITY OPERATIONS DOCUMENTATION

1. Prepare SOC Operations Documentation

1.1 Overview of SOC Tools & Environment

My SOC environment consists of the following core components:

- **Wazuh SIEM (Manager + Indexer + Dashboard)** – central log aggregation, alerting, rules, dashboards.
- **pfSense Firewall** – network perimeter logs, DHCP, packet filter logs.
- **Ubuntu Server (Wazuh Agent)** – system logs, authentication logs, syslog forwarding.
- **UTM Host (macOS)** – virtualization host acting as management plane.
- **SOC Workflow Tools** – ticketing template, escalation flow, response checklist.

Wazuh acts as the **primary SIEM**, receiving logs from:

- pfSense (Firewall logs, DHCP logs, packet filter logs)
- Ubuntu Server (auth logs, sudo logs, system logs, syslog-ng)
- Wazuh internal rules (file integrity monitoring, CIS checks)

1.2 SOC Workflow (with Mermaid Diagram)

Alert Handling Workflow:

flowchart TD

A[Alert Generated in Wazuh] --> B[Alert Categorized: Low/Med/High]
B -->|Low| C[Auto-close with documentation]
B -->|Medium| D[Create SOC Ticket]
B -->|High| E[Escalate to IR Lead]
D --> F[Investigate Logs and Source System]
E --> F
F --> G[Determine True/False Positive]
G -->|True Positive| H[Containment Actions]
G -->|False Positive| I[Close Ticket w/ Notes]
H --> J[Recovery & Hardening]
J --> K[Update Detection Rules]

1.3 Shift-Handover Template

SOC Shift Handover Summary – Example

- **Analyst:** Sid Gautam
 - **Shift:** 08:00–16:00 EST
 - **Open Incidents:**
 - *INC-241:* Suspicious SSH authentication attempts from 10.0.2.15
 - **Pending Escalations:** None
 - **System Health Checks:**
 - Wazuh Manager: Active
 - Indexer: Green
 - Agents: All Connected
 - **Network Alerts:**
 - pfSense packet filter detected outbound DNS spike – investigated, safe.
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2. Document SIEM Implementation

2.1 SIEM Architecture Description

Wazuh SIEM is deployed using Docker Compose:

Architecture Components

- **Wazuh Manager:**
Receives agent events, applies rules, triggers alerts.
- **Wazuh Indexer:**
Elasticsearch-based indexer storing logs and alert metadata.
- **Wazuh Dashboard:**
Web UI for alerts, rules, decoders, agent management.
- **Agents:**
Installed on Ubuntu, pfSense (via syslog), and host services.

2.2 Log Sources (3 Required)

1. Ubuntu Server Logs

- SSH auth events
- sudo failures
- systemd logs
- syslog forwarder logs
Security Value: Detect brute-force attempts, privilege escalation.

2. pfSense Firewall Logs

- Packet filter logs
- DHCP assignments
- Firewall rule triggers
Security Value: Detect scanning, blocked traffic, lateral movement.

3. Wazuh Internal Modules

- File integrity monitoring
 - Vulnerability scanning
 - CIS benchmark checks
Security Value: System misconfigurations, high-risk vulnerabilities.
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2.3 Sample Correlation Rule

Rule Name: Unauthorized SSH Access Attempt

Rule ID: 700010

Logic: Detects > 5 failed SSH auth attempts within 2 minutes.

```
<rule id="700010" level="8">  
  <if_sid>5710</if_sid>  
  <frequency>5</frequency>  
  <timeframe>120</timeframe>  
  <description>Brute-force SSH authentication attempts detected</description>  
  <group>authentication, ssh, brute_force</group>  
</rule>
```

3. Document Threat Detection Principles

3.1 Detection Rule Mechanisms

Wazuh uses:

- **Decoders** – parse raw log fields
 - **Rules** – detect malicious patterns
 - **Groups** – organize alerts
 - **Frequencies/Timeframes** – correlation windows
 - **Thresholding** – control repetition
 - **MITRE ATT&CK Mapping** – classify behaviors
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3.2 Three Detection Scenarios

Scenario 1: SSH Brute Force (Ubuntu)

- Multiple failed authentication attempts
- Rule 700010 triggers
- MITRE Technique: *T1110 – Brute Force*

Scenario 2: Suspicious Port Scan (pfSense)

- pfSense packet filter logs show sequential port hits
- MITRE Technique: *T1046 – Network Scanning*

Scenario 3: File Integrity Violation (Ubuntu)

- `/etc/passwd` modification
- Wazuh FIM triggers

- MITRE Technique: *T1070 – Indicator Removal*
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3.3 Structured Threat Analysis Methodology

1. Identify the event source
 2. Validate authenticity
 3. Correlate with past behavior
 4. Determine threat actor likelihood
 5. Mitigate
 6. Document
 7. Improve detection rules
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4. Document Incident Response Methodology

4.1 Example Incident: SSH Brute Force Attack

Initial Indicators

- Excessive login failures from single IP
 - Wazuh alerts flagged brute-force correlation rule
 - pfSense logs confirm repeated inbound attempts
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4.2 Initial Response Protocol

1. Validate alert integrity
 2. Confirm source (Ubuntu failed logins)
 3. Check successful vs failed attempts
 4. Block malicious IP at pfSense
 5. Force password reset for affected accounts
 6. Document timeline
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4.3 Escalation Criteria

- **Escalate to IR Lead if:**
 - Successful compromise
 - Unauthorized root-level activity
 - Lateral movement detected
 - Privilege escalation logs present
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4.4 Communication Protocols

- SOC Analyst → SOC Lead
- SOC Lead → Security Manager
- If data breach suspected → Legal / Compliance
- If external IP → ISP or law enforcement (if required by policy)

4.5 Case Management Summary (Example)

Ticket ID: INC-241

Incident: SSH authentication brute force

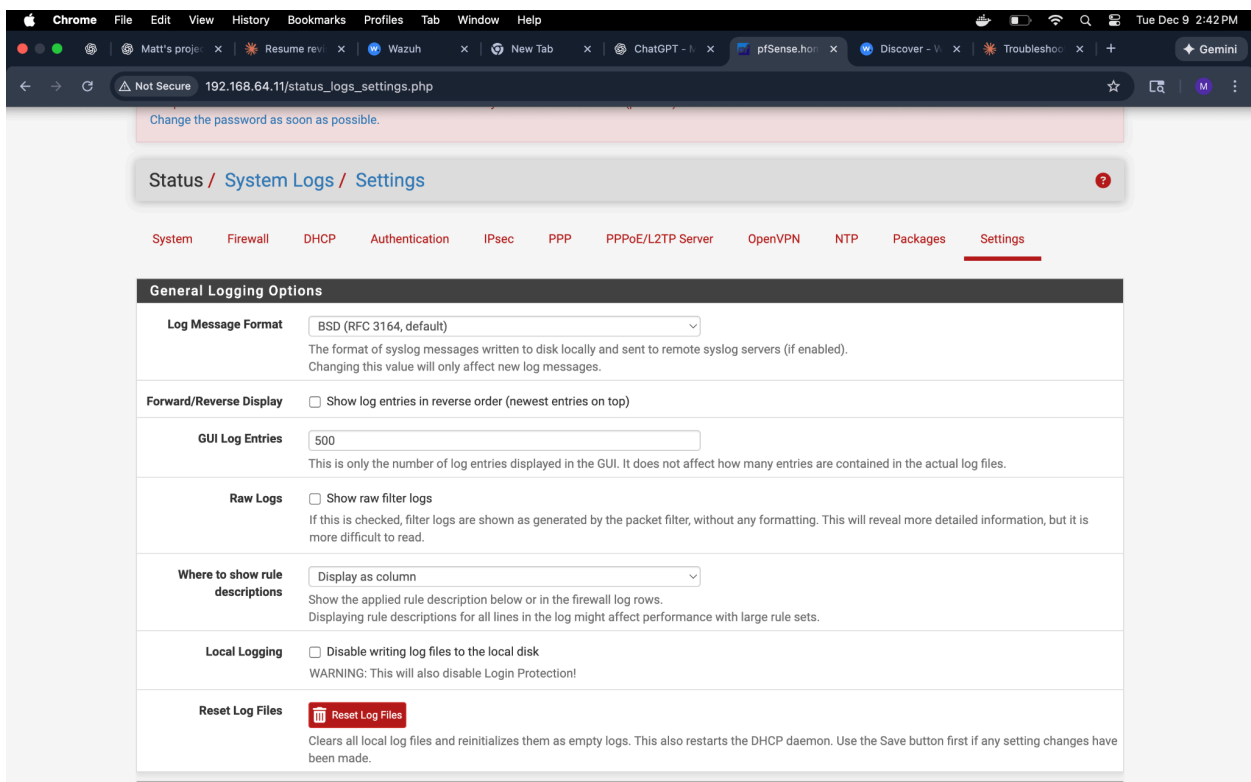
Status: Resolved

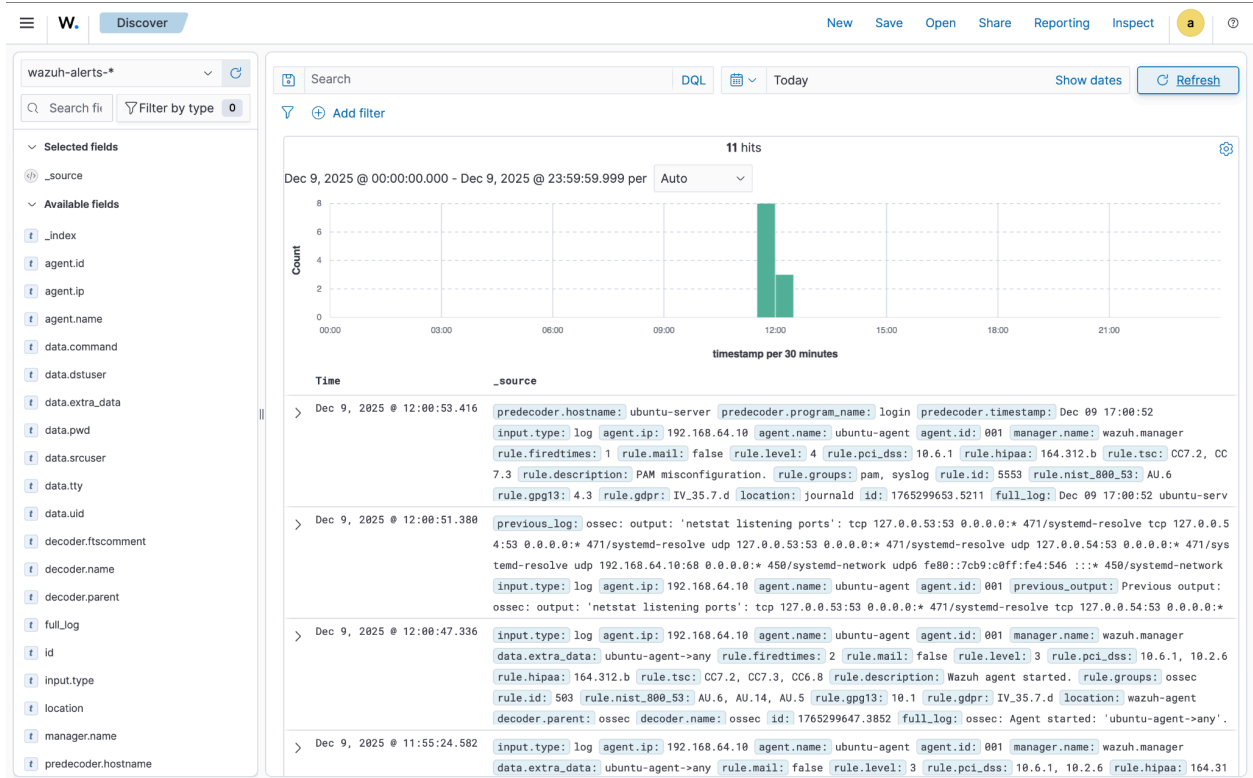
Root Cause: External automated attack

Containment: IP blocked + SSH hardening

Recovery: Updated fail2ban config

Lessons Learned: Add MFA, strengthen monitoring rules





pfSense

Internet6:

Destination	Gateway	Flags	Ne
if Expire			
::/0	fe80::70ae:d5ff:fe23:164%em0	UG	
em0			
::1	link#4	UHS	
lo0			
fd57:cb6d:a6f1:969a::/64	link#1	U	
em0			
fd57:cb6d:a6f1:969a:9478:f6ff:fe18:fcf link#4		UHS	
lo0			
fe80::%em0/64	link#1	U	
em0			
fe80::9478:f6ff:fe18:fcf%lo0	link#4	UHS	
lo0			
fe80::%em1/64	link#2	U	
em1			
fe80::f065:54ff:fe34:7af7%lo0	link#4	UHS	
lo0			
fe80::%lo0/64	link#4	U	
lo0			
fe80::1%lo0	link#4	UHS	
lo0			
[2.8.1-RELEASE][root@pfSense.home.arpa]/root: logger "test pfsense message"			
[2.8.1-RELEASE][root@pfSense.home.arpa]/root:			

